



Medicare DME Policy Assistant

AI-Driven Policy Interpretation for Billing and Coding Compliance

Cotiviti Intern Assessment - Topic 3: Content Management in Health Care

3 CMS Policy Documents

Real Medicare billing rules

558 Passages Indexed

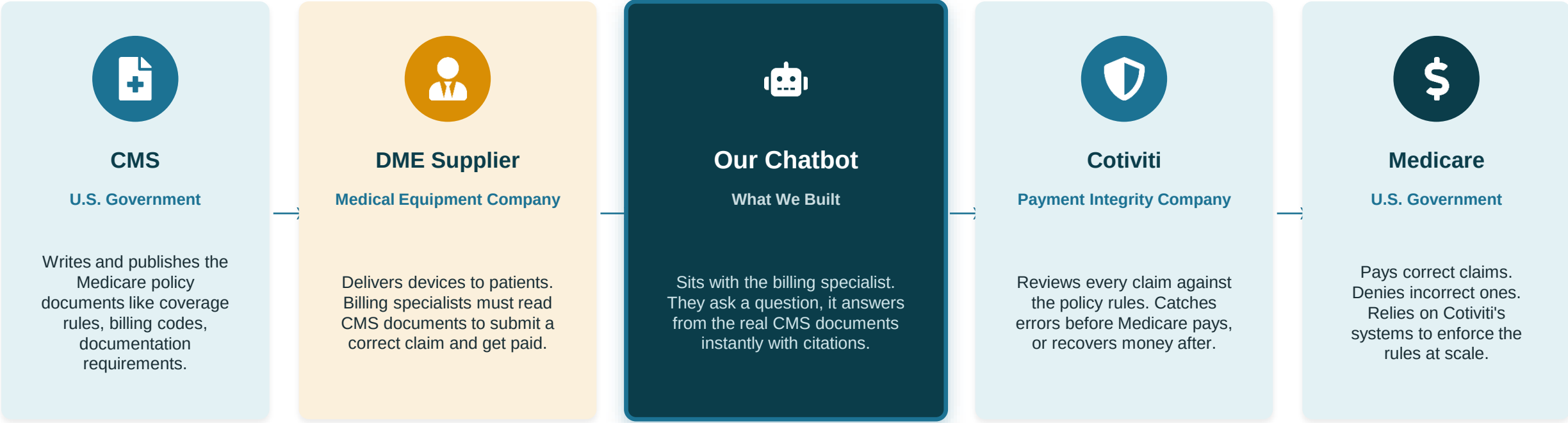
Instantly searchable

98.2% Accuracy

55-question validation

Before Anything Else: Who Is Who?

Understanding where Cotiviti sits in the Medicare payment chain makes everything else clear.



The chatbot lives with the supplier's billing specialist, **not inside Cotiviti**. When the specialist submits a correct claim, Cotiviti's systems have less to catch. Better claims going in means better payment integrity coming out.

The Market Cotiviti Operates In: \$2.3 Billion in Preventable Errors

Every dollar in this chart is either Cotiviti’s existing revenue opportunity, or a new one if this chatbot exists.



\$28.8B

Medicare Improper Payments (FY2025)

6.55% of all Medicare fee-for-service spend paid incorrectly (CMS Fact Sheet, Jan 2026)



24.1%

Error Rate in DME Claims

\$2.3B in DME errors, the exact category this chatbot addresses (CMS FY2025 Supplemental Data)



64%

Errors from Documentation Gaps

Not fraud, just wrong codes, missing paperwork, misapplied policy. Fully preventable with better tools.

TODAY: Cotiviti catches errors AFTER claims are submitted. They already earn revenue doing this.

WITH THIS CHATBOT: Cotiviti can also earn revenue PREVENTING errors at the source, before claims are submitted.

The Solution: Ask a Question, Get a Cited Answer

Instead of manually searching 3 documents, billing teams type a plain-English question and get an instant, sourced response.

What It Does



Understands plain English questions about Medicare billing policy



Searches 3 official CMS policy documents instantly



Returns a precise answer grounded only in real policy text



Shows exactly which document and section the answer came from



Blocks off-topic or irrelevant questions automatically

The Three Rulebooks It Knows



Coverage Rules

LCD L33822

Who qualifies for a glucose monitor and under what medical conditions?



Billing Codes

Policy Article A52464

Which billing code and modifiers to use, and what can be billed together?



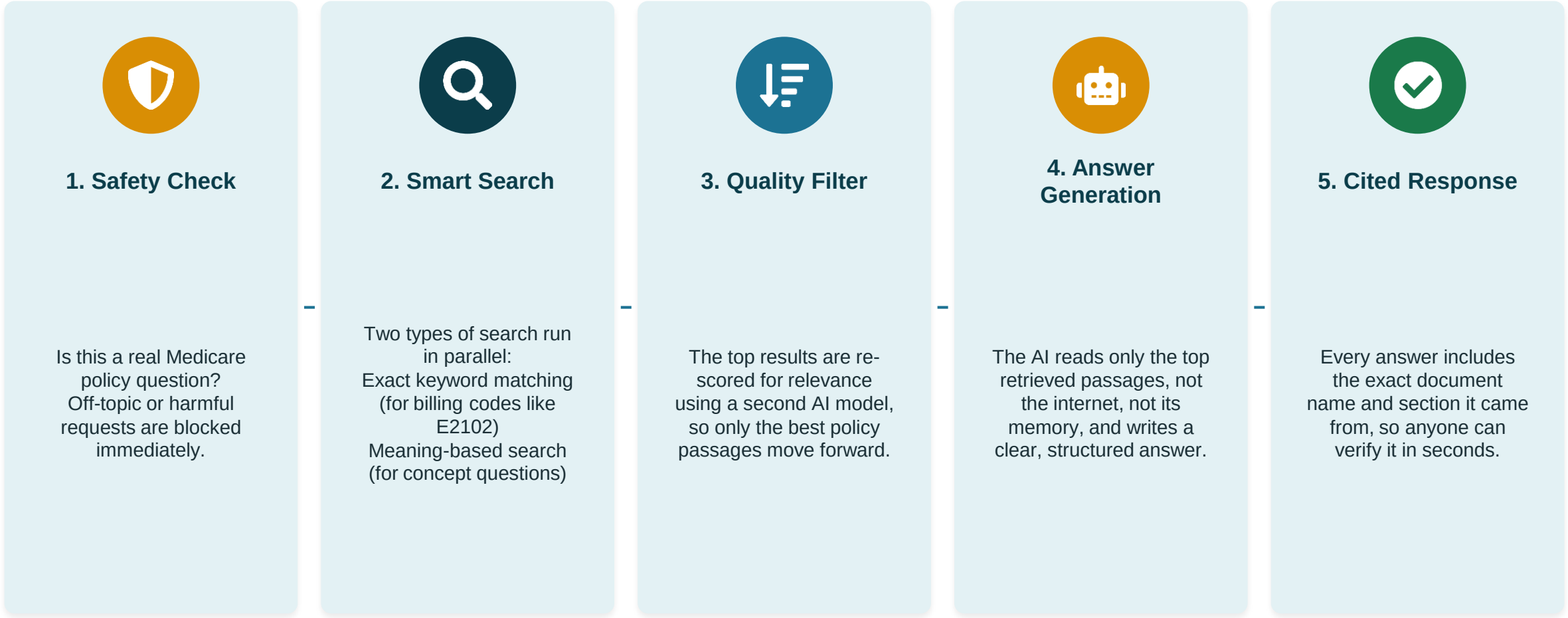
Documentation Rules

Article A55426

What paperwork must exist, orders, delivery proof, refill records?

How It Works: 5 Steps From Question to Answer

No jargon. Here is exactly what happens when a billing specialist types a question.



Under the Hood: Technology Stack

For technical reviewers - production-grade components, not demo shortcuts.

Backend	Python FastAPI
Text Search	BM25Okapi - exact keyword matching (catches billing codes like A4238, E2102)
Semantic Search	FAISS vector index with all-MiniLM-L6-v2 embeddings - meaning-based similarity
Reranker	cross-encoder/ms-marco-MiniLM-L-6-v2 - re-scores top results with full query context
Language Model	Gemini 2.5 Flash (Google) - generates answer only from retrieved passages
Frontend	HTML / CSS / JavaScript - browser-based UI, served by FastAPI
Corpus	3 CMS policy documents 558 passages indexed Hybrid retrieval index
Guardrails	Input validation - topic scoping, injection detection, off-topic rejection

The Chatbot in Action

• • • Medicare DME Policy Assistant - Cotiviti POC v2.0

Medicare DME
Policy Assistant
Cotiviti POC - v2.0

● API key loaded from server

KNOWLEDGE BASE

- LCD L33822 – Glucose Monitors
- Policy Article A52464
- Standard Documentation A55426

RAG PIPELINE

1. Guardrails & topic scoping
2. Query classification
3. BM25 keyword search
4. FAISS semantic search
5. Reciprocal Rank Fusion
6. Cross-encoder re-ranking
7. Groq LLM generation
8. Confidence scoring

MODELS

Embed: all-MiniLM-L6-v2
Rerank: cross-encoder/ms-marco-MiniLM-L-6-v2
LLM: gemini-2.5-flash

Medicare Glucose Monitor Coverage Q&A

567 passages indexed

tell me about glucose monitors

A

Based on the provided policy excerpts, here is information about glucose monitors:

I. Types of Glucose Monitors

The policy documents refer to two main types of glucose monitors:

Home Blood Glucose Monitors (BGM)

(According to LCD L33822 – Glucose Monitors | Section: HOME BLOOD GLUCOSE MONITORS (BGM))

Continuous Glucose Monitors (CGM), which can be non-adjunctive or adjunctive (According to LCD L33822 – Glucose Monitors | Section: HOME BLOOD GLUCOSE MONITORS (BGM) and Policy Article A52464 – Glucose Monitor | Section: MODIFIERS)

II. Classification of Continuous Glucose Monitors (CGM)

On February 28, 2022, CMS determined that both non-adjunctive and adjunctive CGMs may be classified as Durable Medical Equipment (DME). (According to LCD L33822 – Glucose Monitors | Section: HOME BLOOD GLUCOSE MONITORS (BGM))

III. HCPCS Codes for Glucose Monitors and Related Supplies

The following HCPCS codes are mentioned:

Blood Glucose Monitors: E0607, E2100, E2101, E2104 (According to Policy Article A52464 – Glucose Monitor | Section: MODIFIERS)

CGM Devices:

E2102 (for adjunctive CGM incorporated into an insulin infusion pump) (According to Policy Article A52464 – Glucose Monitor | Section: MODIFIERS)

E2103 (for non-adjunctive CGM device) (According to LCD L33822 – Glucose Monitors | Section: MEDICINE CESSITY and Policy Article A52464 – Glucose Monitor | Section: MODIFIERS)

Clear Chat

Ask about Medicare glucose monitor coverage...

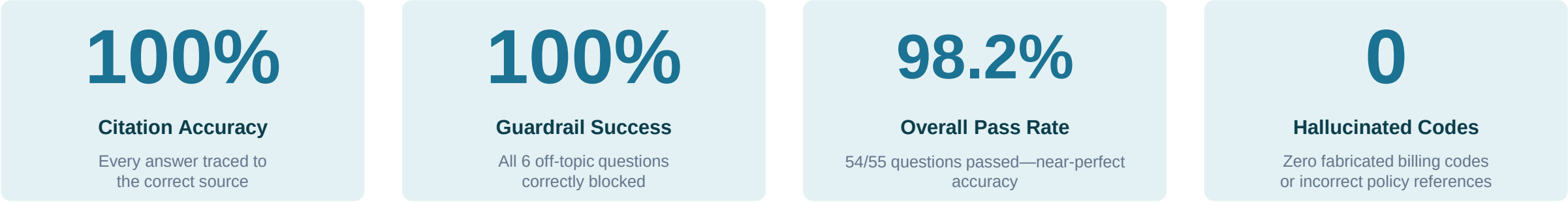
Send

► Try it live: chatbot-cotiviti.streamlit.app - visitors enter their own free Gemini API key in the sidebar

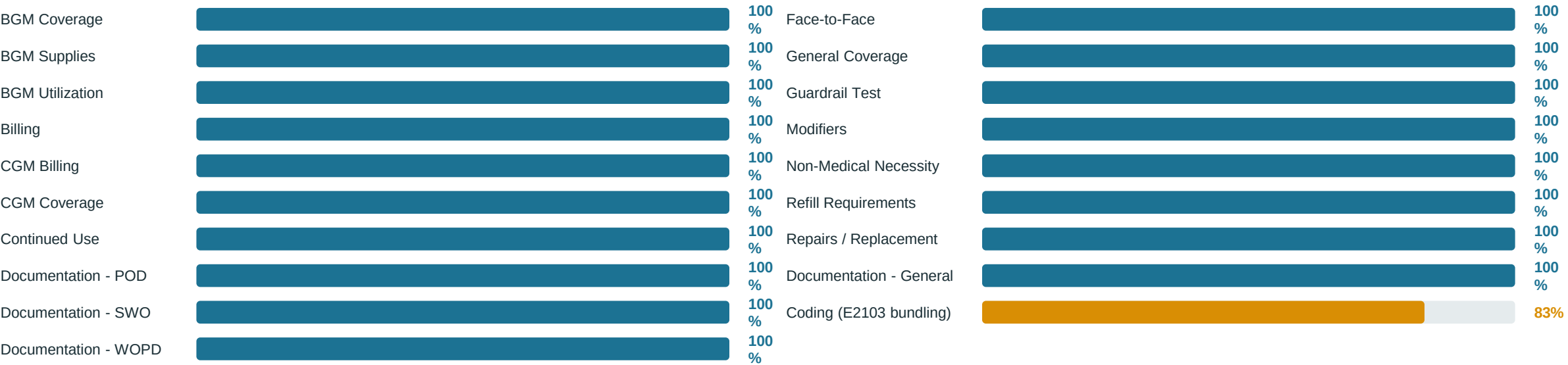
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Does It Actually Work? Validation Results

55 questions written by hand across 19 categories - tested against the real CMS policy documents.



Category Breakdown: 18 of 19 at 100%



Only weak spot: Coding (83%) - E2103 Column I/II bundling question missed 2 keyword variants (E0607, A4253) even though citations were still correct.

Opportunities for Cotiviti

The same system generalizes far beyond glucose monitors, to any coverage policy, any payer contract.



Faster Policy Lookup for Billing Teams

Coding and billing staff get precise, cited answers in seconds instead of 20-minute manual searches—directly reducing claim errors and denial rates.



Automatic Policy Change Alerts

When CMS revises a coverage rule, the system can detect and summarize exactly what changed—giving compliance teams advance notice before the effective date.



Pre-Submission Claim Screening

Check claims against coverage criteria before they go to the payer, flagging likely denials before submission, not after. Every caught error saves recovery cost.



Contract Intelligence

The same architecture can index payer-provider contracts like fee schedules, carve-outs, and policy references to resolve payment disputes and surface underpayments.

Under the Hood: Why Each Technical Choice Was Made

Every component was chosen for a specific reason, not convenience.

Retrieval Pipeline

BM25 Keyword Search

Why: Billing policy contains exact codes (E2102, A4238, KF modifier) that semantic search alone will miss if phrasing differs slightly. BM25 anchors precision to the exact token, no guessing.

FAISS Semantic Search

Why: Handles paraphrased or conceptual queries (“what paperwork do I need?”) where keyword matching returns nothing. Converts meaning, not just words, into a numeric fingerprint for comparison.

Reciprocal Rank Fusion

Why: Merges the two ranked lists without needing to normalize scores across different scales. Each result is scored by its rank position, not raw similarity values, simple and robust.

Cross-Encoder Reranking

Why: FAISS bi-encoder similarity is computed without seeing the query and passage together, fast but approximate. The cross-encoder reads both simultaneously, giving far more accurate relevance for the top results.

Generation & Grounding

Temperature = 0

Why: Compliance-adjacent tool: the same question must return the same answer every time. A hallucinated billing code is not a chatbot quirk, it’s a coding error with real financial consequences.

Grounded-Only System Prompt

Why: Gemini is explicitly instructed to answer only from the retrieved passages, never its training knowledge. Medicare policy changes frequently; training knowledge goes stale silently.

Section-Aware Chunking

Why: CMS documents already use semantic section headers (e.g. “MODIFIERS”, “COVERAGE CRITERIA”). Splitting on those boundaries keeps each chunk topically whole rather than cutting a rule mid-sentence.

Confidence Scoring

Why: Not all questions have a clean answer in the corpus. Low-confidence responses are flagged for human review rather than served as fact, a key guardrail before any production deployment.

Risks and How to Address Them

Each is real. Each has a concrete counter. None is a reason not to build it.

⚠ Risk

The AI Could Still Make Mistakes

Even with retrieved passages, the AI may give an imprecise or incomplete answer, especially on complex multi-condition queries.

✓ How to Fix It

Confidence scoring flags low-certainty answers for human review. All answers show their sources so a specialist can verify before acting.

⚠ Risk

Policy Documents Change Frequently

CMS revises LCDs and policy articles dozens of times a year. An outdated index silently gives wrong answers.

✓ How to Fix It

Build a re-ingestion pipeline triggered by CMS revision dates. Add revision date labels to every retrieved passage so users see when it was last current.

⚠ Risk

Regulatory and Liability Exposure

AI-assisted coverage interpretation could be construed as legal or compliance advice, especially under new state AI-disclosure laws.

✓ How to Fix It

Clear disclaimer framing (“research aid, not a coverage determination”), audit logging of every query and response, and human escalation paths before deployment.

Strategic Recommendations for Cotiviti

Three concrete actions in priority order, each builds on the last.

1

Build an Internal RAG Platform for Policy Documents

Start with DMEPOS LCDs (where the 24.1% improper payment rate is highest), then expand to all NCDs and payer contracts. Give internal coding teams a citation-grounded Q&A interface that keeps up with CMS revision dates automatically.

2

Launch a Policy Change Monitoring Service

Use AI to detect and summarize changes between LCD versions, delivering proactive alerts to payer and provider clients before new rules take effect. In a market where regulatory velocity is accelerating, early detection is a defensible, high-value differentiator.

3

Establish an AI Governance Framework First

Define confidence thresholds, human review escalation paths, source attribution requirements, and audit logging standards before broader deployment. Cotiviti's credibility in payment integrity depends on AI recommendations being explainable and traceable, not just accurate in testing.

How This Grows Cotiviti's Business

This chatbot is not just a tool, it's a new revenue stream and a stronger core product at the same time.

Cotiviti currently earns revenue catching errors AFTER claims are submitted. This chatbot creates a second revenue model: earning revenue PREVENTING errors BEFORE claims are submitted, on the supplier side.



New Revenue Stream

Sell to DME Suppliers

Thousands of DME suppliers across the U.S. all have this problem. None have a good tool for it today. Cotiviti can sell a subscription-based policy Q&A tool to the very companies whose claims they already audit, creating a new product line without new infrastructure.



Better Core Product

Cleaner Claims = Better Audits

When suppliers submit correct claims from the start, Cotiviti's audit systems can focus on true anomalies and fraud instead of administrative errors. Their existing payment integrity product becomes more accurate and more valuable.



Unique Competitive Moat

Only Cotiviti Can Build This Version

Cotiviti has years of data on how claims go wrong, which codes get confused, which rules get missed most often, which DME categories have the highest denial rates. No competitor can add those warnings. Their version of this tool is fundamentally better than anyone else's.



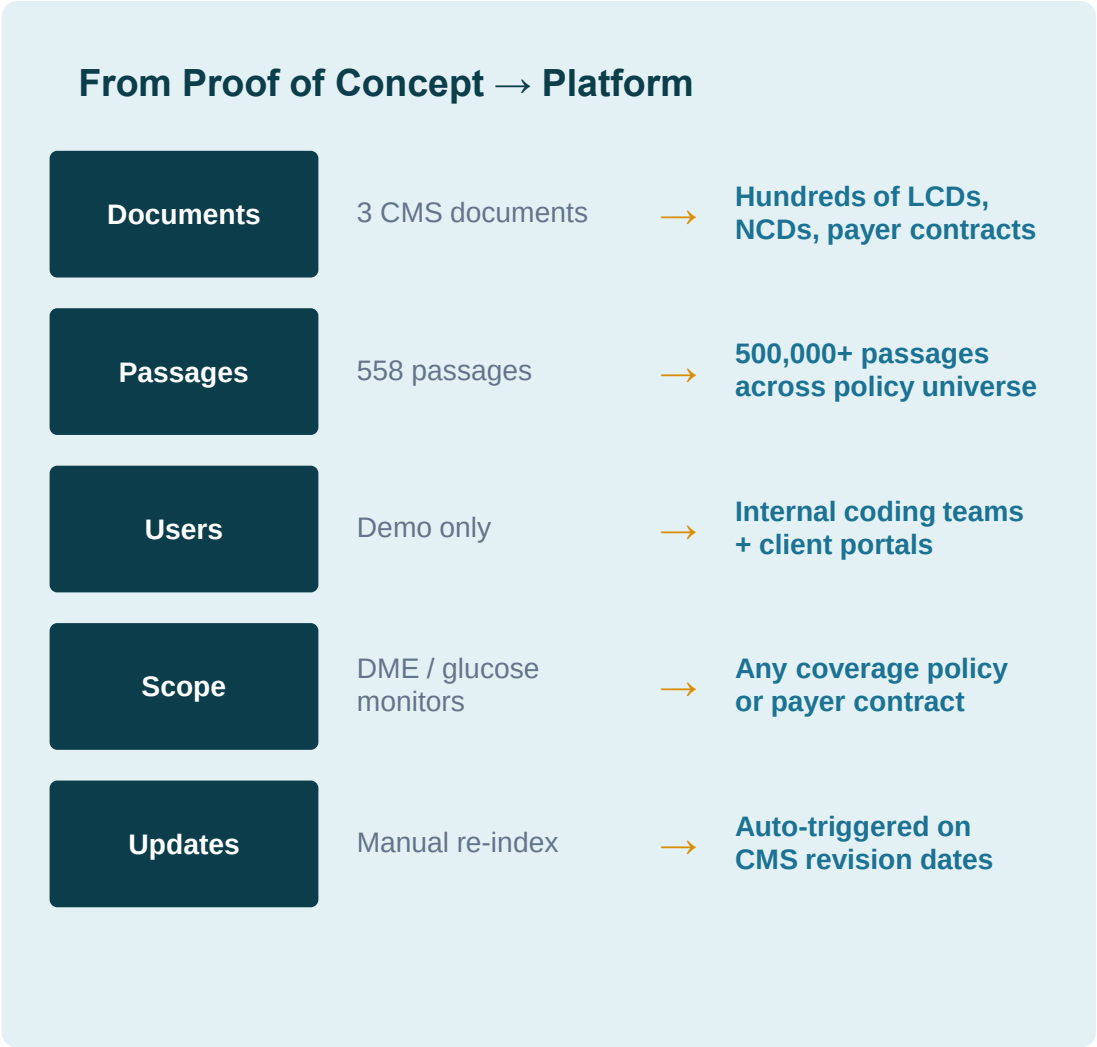
Platform That Compounds

Start Small, Scale Fast

Glucose monitors today → all DMEPOS → all LCDs → payer contracts. Every new document type added is a new upsell to existing customers, not a new product build. The architecture already supports it, it's an ingestion step, not a rebuild.

Business Impact at Scale

From a 3-document glucose-monitor proof of concept to a platform covering every policy Cotiviti touches.





Faster Policy Lookup

Minutes of manual research compressed to seconds—per analyst, per day, at scale.



Fewer Claim Denials

Catch coding errors before submission. A prevented denial costs nothing to recover.



Reusable Platform

Same architecture works for any LCD, NCD, or contract—no rebuild required.

Why This Advantage Compounds Over Time

The model is the easy part. What's hard to replicate is the corpus, the eval suite, and the domain expertise.



Domain-Specific Policy Knowledge

Anyone can connect an AI to a document. Building a corpus that correctly handles the coverage rule / billing code / documentation split, and stays current with CMS revisions, requires healthcare policy expertise, not just engineering. That expertise is Cotiviti's core asset.



Growing Evaluation Infrastructure

The 55-question test suite built for this POC is the seed of a regression framework. As the corpus grows, so does the eval suite. A competitor starting from zero has no benchmarks, no failure modes on record, and no baseline to improve against.



Auditability as the Differentiator

In payment integrity, a black-box answer has no value. Every response must show the source document, the section, and the revision date, so auditors and compliance teams can verify it. Consumer AI tools can't offer this. It's the exact design principle here.



Platform Vision

Glucose monitors → all DMEPOS → all LCDs → all NCDs → all payer contracts. Same architecture, same eval framework, same interface. Built once, extended indefinitely.

Current Limitations and How to Improve Them

Honest about what this proof of concept doesn't do yet, and the clear path to fix each gap.

Current Limitation

Only covers 3 CMS documents on one device type
Validation uses keyword matching, not human expert review
No automatic updates when CMS revises a policy
Each question is independent, no conversation memory
Not security-hardened or HIPAA-reviewed for production
Confidence scoring is rule-based, not statistically calibrated

Path to Improvement

Expand corpus to all DMEPOS LCDs, then NCDs, then payer contracts. Architecture already supports this with no changes.
Bring in a Medicare billing specialist to review a sample of answers and build a human-judged benchmark.
Build a lightweight monitoring job that checks CMS revision dates and triggers re-ingestion automatically.
Add multi-turn dialogue so users can follow up ("What if the patient is non-insulin-treated?") without re-asking from scratch.
Conduct a security review and HIPAA assessment before deploying with any real patient or claims data.
Calibrate confidence thresholds against human expert judgments so the system knows when to escalate vs. answer.



Why This Fits Cotiviti

*Cotiviti's payment integrity business already "codifies CMS and AMA guidelines into rules."
This system speeds up the research step in front of that work, with citations, not a black box.*

100% citation accuracy

Zero hallucinated codes

Scales to any LCD, NCD, or contract

chatbot-cotiviti.streamlit.app

Live demo , visitors use their own free Gemini API key

GitHub: github.com/HariPrakash79/cotiviti-medicare-chatbot